concerns and to avoid excessive noise pollution. As a result, 150 firm orders for the Airbus were on the books even before the aircraft's maiden flight.

Boeing fought back strongly, if somewhat belatedly, with a longer-range version of the existing 777 and, above all, with the 787 Dreamliner, which entered service in 2011. Despite its kitsch name, the Dreamliner constituted an intelligently pitched answer to the Airbus A380. As a mid-size airliner, it was able to fly direct into the smaller airports that the A380 could not use, obviating the need for transfer flights. Making extensive use of lightweight materials and aluminum alloys, it promised ultra-efficient fuel use and maintenance, quiet engines, and a range of 9,775 miles (15,700km).

**AIRBUS ASSEMBLY**
*The A380 superjumbo comes together on the Airbus assembly line at Toulouse, surrounded by a complex structure of gantries. The enormous assembly hall is one of the largest buildings in the world, covering the area of 20 football fields. The hall's construction required four times as much steel as the Eiffel Tower.*

## Global manufacture

The head-to-head between Airbus and Boeing obviously had aspects of an old-style contest between national air industries. Boeing's vociferous complaints about alleged unfair support for Airbus from European governments were countered by allegations of hidden US subsidies to Boeing. Yet both the Airbus A380 and the Boeing 787 demonstrated the impossibility of the pursuit of narrow nationalism in the 21st-century global economy. International cooperation was, of course, basic to Airbus Industrie, which prided itself on being a joint European enterprise, but the 787 was also far from being a specifically American piece of work. The Japanese companies Fuji and Kawasaki were major contributors to the aircraft, manufacturing much of the fuselage and wings, and Korean, British, and French companies were among those involved in the project.

## Environmental challenge

Ultimately the greatest threat to the future expansion of global air travel probably lies in the rising sensitivity to the environmental damage caused by aircraft. In the course of the 21st century, it seemed as though those governments committed to combating global warming were bound eventually to listen to their scientific advisers telling them that airliners were a large part of the problem of excessive emission of "greenhouse gases." Airlines and aircraft manufacturers were naturally hostile to the introduction of "green" taxes as a means of increasing the cost of air travel and thereby deterring people from making unnecessary flights. On the other hand, they liked the idea of of fuel efficiency, which would translate not only into lower pollution levels but also into lower operating costs and therefore greater profitability. Both the Boeing Dreamliner and Airbus A380 had fuel efficiency as a key selling point. However, environmental campaigners remained unimpressed. If fuel efficiency were to be used to make flying still cheaper, then it would only lead to more flights rather than a cleaner atmosphere. These issues continue to pose a stimulating challenge to those involved in the development of future patterns of passenger air travel and commercial aircraft technology.

**ICE AND SNOW**
*Ability to operate safely in cold weather conditions is an essential requirement for any modern airliner. Here the A380 is put to the test in extreme snow and ice. The exhaustive flight test process required around 2,500 hours of flying by four A380s carried out over a period of more than a year.*


**LANDING ON WATER**
*An Airbus A380 makes high-speed passes through a water-filled trough on the runway to check that spray from the wheels does not impair engine operation.*